

Algo-Trading Pipeline

SLIBRARY 24 · SHEET 1 / 5

Signal → execution → monitoring

PROFESSIONAL DEFINITION

The Algorithmic Trading Pipeline describes the flow from signal generation through execution to monitoring, showing how automated strategies translate data into trades and manage their consequences.

WHO DEVELOPED IT

A construct from quantitative-finance and trading-systems practice.

WHY IT WAS DEVELOPED

It was developed to structure the design and governance of automated trading, where errors propagate at machine speed.

FIGURE 1 · ALGO-TRADING PIPELINE



Figure 1. A schematic of the Algo-Trading Pipeline as applied in BARBM312 (illustrative).

HOW IT IS USED

Each stage — signal, execution, monitoring — is designed with controls; the pipeline clarifies where risk and oversight must sit.

WHEN IT IS USED

When designing or governing algorithmic trading systems.

BY WHOM IT IS USED

Quantitative traders, engineers and risk teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It makes the end-to-end automated trading process explicit and governable, emphasising monitoring as integral, not optional.

RELATED FRAMEWORKS IN THIS COURSE

Latency Arms Race · Slibrary 24

Market Microstructure Map · Slibrary 24

Flash-Crash Risk Loop · Slibrary 24

Execution Cost Frontier · Slibrary 24

SOURCES (HARVARD REFERENCING STYLE)

Chan, E.P. (2013) Algorithmic Trading: Winning Strategies and Their Rationale. Hoboken: Wiley.

Aldridge, I. (2013) High-Frequency Trading. Hoboken: Wiley.